

ANSWER KEY

Grade 12 GS: Advanced Mechanical Oscillations

- 1. Answer: (b)** $\theta'' + \frac{Mga}{I}\theta = 0$
Rationale: This equation correctly relates the restoring torque from gravity (Mga) and the moment of inertia (I) to the angular acceleration.
- 2. Answer: (c)** The distance from the pivot point to the center of mass G .
Rationale: The gravitational torque depends on the lever arm distance from the axis of rotation to the point where gravity effectively acts (G).
- 3. Answer: (c)** $T' = \frac{T}{2}$
Rationale: Since $T \propto \frac{1}{\sqrt{C}}$, quadrupling C results in dividing the period by $\sqrt{4} = 2$.
- 4. Answer: (b)** To ensure the restoring force is linear (Linearization).
Rationale: The derivation relies on $\sin \theta \approx \theta$. Large angles make the restoring force non-linear, leading to anharmonic motion.
- 5. Answer: (b)** It remains unchanged.
Rationale: In a simple pendulum, the mass cancels out because both the inertial term and the gravitational restoring force are proportional to mass.
- 6. Answer: (b)** Elastic Potential Energy converts to Kinetic Energy.
Rationale: At maximum displacement, the energy is stored in the twist of the wire. As it unwinds, this potential energy accelerates the disk.
- 7. Answer: (b)** Use the Parallel Axis Theorem: $I = \frac{1}{12}ML^2 + M(\frac{L}{2})^2$.
Rationale: You must shift the moment of inertia from the center of mass to the pivot point located distance $L/2$ away.
- 8. Answer: (b)** The motion is damped by non-conservative forces.
Rationale: A decrease in total mechanical energy indicates work is being done by resistive forces like air friction or internal friction.
- 9. Answer: (b)** Torsion Constant (C)
Rationale: Just as $F = -kx$ describes a spring, $\tau = -C\theta$ describes a torsion wire, making C the stiffness analog.
- 10. Answer: (b)** The period approaches infinity.
Rationale: As a approaches 0, the restoring torque ($Mga \sin \theta$) approaches 0. Without a restoring torque, the object will not oscillate (infinite period).